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IMPELLER WITH VARIABLE BACKSWEEP ANGLE

Abstract

Described herein is an impeller for a flow device. The impeller comprises a hub, and a plurality of blades extending radially outwards from the hub, wherein each of the blades is at least partially attached to an outer surface of the hub such that at least a portion of a tip end, opposite to the hub, of the corresponding blades remains unattached to the outer surface of the hub, wherein the unattached portion of the blades is free to deflect based on a rotational speed of the impeller and correspondingly adjust a backsweep angle of the corresponding blades within a predefined range.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This patent application claims the benefit of U.S. Provisional Patent Application No. 63/563,535, filed on Mar. 11, 2024, which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] The subject disclosure relates to the field of impellers, and more particularly, to an impeller with variable backsweep angle.

SUMMARY

[0003] Described herein is an impeller for a flow device. The impeller comprises a hub, and a plurality of blades extending radially outwards from the hub, wherein each of the blades is at least partially attached to an outer surface of the hub such that at least a portion of a tip end, opposite to the hub, of the corresponding blades remains unattached to the outer surface of the hub, wherein the unattached portion of the blades is free to deflect based on a rotational speed of the impeller and correspondingly adjust a backsweep angle of the corresponding blades within a predefined range.

[0004] In one or more embodiments, the unattached portion of the blades extends at least partially beyond an outermost diameter of the hub.

[0005] In one or more embodiments, the predefined range of the backsweep angle ranges from 15 to 60 degrees from a radial direction from the hub.

[0006] In one or more embodiments, each of the blades has a substantially curved profile, extending in a predefined direction about a longitudinal axis or axis of rotation of the hub, such that a plurality of fluidic passages is formed or defined between the adjacent blades to allow flow of a fluid therethrough.

[0007] In one or more embodiments, the outer surface of the hub has a substantially curved profile, wherein an outer diameter of the hub increases in a direction while moving from a leading end towards a trailing end of the hub, wherein the unattached portion of the hub extends at least partially beyond the outer diameter of the trailing end of the hub.

[0008] In one or more embodiments, each of the blades comprises a coupling side attached to the outer surface of the hub, a leading edge located adjacent to a leading end of the hub, a trailing edge located adjacent to a trailing end of the hub, and a curved edge opposite to the coupling side and extending between the trailing edge and the leading edge of the corresponding blades.

[0009] In one or more embodiments, the trailing edge of the plurality of blades are circumferentially offset from the corresponding leading edge of the blade.

[0010] In one or more embodiments, a portion, at the leading edge, of each of the blades is substantially curved in a direction opposite to the predefined direction.

[0011] In one or more embodiments, each of the blades has a variable thickness between the leading edge and the trailing edge of the corresponding blades.

[0012] In one or more embodiments, the thickness of each of the blades reduces while moving in a radial direction from the coupling side towards the tip end or outer edge of the corresponding blades.

[0013] In one or more embodiments, each of the blades has a variable thickness between the coupling side and the curved edge of the corresponding blades.

[0014] In one or more embodiments, the thickness of each of the blades reduces while moving in a direction from the leading edge towards the trailing edge of the corresponding blades.

[0015] In one or more embodiments, the thickness of each of the blades increases while moving from the leading edge towards a middle portion of the corresponding blades and further reduces while moving towards the trailing edge of the corresponding blades.

[0016] In one or more embodiments, the impeller comprises a shroud encasing the hub and the

plurality of blades, such that a first end of the shroud located adjacent to the leading edge of the blades remains open to allow inflow of a fluid therewithin, and a second end of the shroud located adjacent to the trailing edge of the blades remains open to allow outflow of the fluid therefrom.

[0017] In one or more embodiments, the shroud encases the hub and the plurality of blades, such that the unattached portion of the blades remains enclosed by and unattached to the shroud.

[0018] In one or more embodiments, the shroud has a substantially curved profile, wherein a diameter of the shroud increases while moving in a direction from a leading end towards a trailing end of the hub.

[0019] In one or more embodiments, the plurality of blades, the hub, and the shroud are made of a non-metallic material.

[0020] In one or more embodiments, the plurality of blades is made of a composite material having a predetermined flex modulus.

[0021] In one or more embodiments, the impeller comprises a plurality of splitters configured in an alternating arrangement with the plurality of blades such that one of the splitters remains between adjacent blades, wherein the splitters extend radially outwards from the hub such that at least a portion of a tip end, opposite to the hub, of the corresponding splitter remains unattached to the outer surface of the hub.

[0022] In one or more embodiments, the flow device is selected from a group comprising a centrifugal compressor, a centrifugal pump, and a turbine.

[0023] The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, features, and techniques of the subject disclosure will become more apparent from the following description taken in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] The accompanying drawings are included to provide a further understanding of the subject disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the subject disclosure and, together with the description, serve to explain the principles of the subject disclosure.

[0025] In the drawings, similar components and/or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label with a second label that distinguishes among the similar components. If only the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

[0026] FIGS. 1A to 1C illustrate exemplary views of an unshrouded impeller for a flow device (centrifugal compressor) in accordance with one or more embodiments of the subject disclosure.

[0027] FIGS. 2A to 2C illustrate exemplary views of a shrouded impeller for a flow device (centrifugal compressor) in accordance with one or more embodiments of the subject disclosure.

[0028] FIGS. 3A and 3B illustrate exemplary top-view representations of the impeller to depict the variation in backsweep angle at different rotational speeds in accordance with one or more embodiments of the subject disclosure

DETAILED DESCRIPTION

[0029] The following is a detailed description of embodiments of the subject disclosure depicted in the accompanying drawings. The embodiments are in such detail as to clearly communicate the subject disclosure. However, the amount of detail offered is not intended to limit the anticipated variations of embodiments; on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the subject disclosure as defined by the

appended claims.

[0030] Various terms are used herein. To the extent a term used in a claim is not defined below, it should be given the broadest definition persons in the pertinent art have given that term as reflected in printed publications and issued patents at the time of filing.

[0031] The terms “leading end”, as used herein, refer to an upstream end of the impeller, which first receives or comes in contact with the fluid flowing through the flow device, regardless of whether the flow device is a compressor, pump, or turbine. Further, the terms “trailing end”, as used herein, refer to a downstream end of the impeller, which is opposite to the upstream end.

[0032] The terms “longitudinal axis” and “axis of rotation”, as used herein, refer to the z-axis, about which the impeller in the flow device rotates, regardless of whether the flow device is a compressor, pump, or turbine. Further, the terms “radial direction”, as used herein, refer to an axis that is normal to the z-axis.

[0033] In the specification, reference may be made to the spatial relationships between various components and to the spatial orientation of various aspects of components as the devices are depicted in the attached drawings. However, as will be recognized by those skilled in the art after a complete reading of the subject disclosure, the components of the subject disclosure, described herein may be positioned in any desired orientation. Thus, the use of terms such as “above,” “below,” “upper,” “lower,” “first”, “second” or other like terms to describe a spatial relationship between various components or to describe the spatial orientation of aspects of such components should be understood to describe a relative relationship between the components or a spatial orientation of aspects of such components, respectively, described herein may be oriented in any desired direction.

[0034] Flow devices such as but not limited to centrifugal compressors play an important role in a wide array of applications, such as in HVAC systems. The efficiency, performance, and operational flexibility of these compressors may be significantly influenced by the design of their impellers, particularly the geometry of the impeller blades. The backsweep angle of impeller blades is an important design parameter that may affect the performance characteristics of centrifugal compressors. This backsweep angle, defined by the orientation of the blade's trailing edge relative to the direction of rotation, may influence the compressor's operating range, efficiency, and the head it can produce.

[0035] Typically, an impeller with a larger backsweep angle may have a wider operating range but may sacrifice some head, making it suitable for applications requiring varied flow conditions. Conversely, an impeller with a smaller backsweep angle may generate a higher head, beneficial for applications where a significant elevation of pressure is needed at a relatively constant flow rate.

[0036] The operating points of chillers and heat pumps (HPs), which are important components in refrigeration and heating systems, respectively, may often necessitate specific performance characteristics from the centrifugal compressors they involve. Adjusting the backsweep angle of the impeller blades may optimize the compressor's performance to match the required operating point, enhancing the overall efficiency and effectiveness of the system.

[0037] However, conventional impeller designs, both shrouded and unshrouded, have fixed backsweep angles. These blades, stretching from the leading edge to the trailing edge, are rigidly affixed to the hub (and to the shroud in the case of shrouded impellers). As a result, once manufactured, the backsweep angle of these impellers cannot be adjusted to cater to varying operational demands or to optimize performance across a broader range of conditions. This inherent inflexibility may restrict the adaptability of centrifugal compressors, compromising either on the efficiency or operational range, depending on the fixed geometry of the impeller blades.

[0038] There is, therefore, a need for a solution to effectively address the challenges associated with existing impellers used in flow devices (centrifugal compressors), by providing an improved impeller capable of adjusting its backsweep angle in response to changing operational requirements. This may improve the overall performance and operating range of the flow devices

equipped with this impeller.

[0039] Referring to FIGS. 1A to 2C, an impeller **100** for use with a flow device associated with a heating, ventilation, and air conditioning (HVAC) system (not shown) is disclosed. In one or more embodiments, the flow device can be selected from a group comprising a centrifugal compressor, a centrifugal pump, and a turbine, but is not limited to the like. While various embodiments have been described herein for the flow device being a centrifugal compressor for the sake of brevity, however, the impeller may also be associated with other flow devices as well without any limitation.

[0040] In one or more embodiments, the impeller **100** can include a hub **102** having an outer surface **102A** (along a flow path of fluid in the flow device) extending from a leading end **102-1** (upstream end) **102-1** towards a trailing end **102-2** (downstream end) **102-2** of the hub **102**. The hub **102** can further include an inner diameter surface **102-B** at one end which can be configured to receive and mate with a shaft of a drive (motor and/or gears) (not shown) associated with the flow device, such that rotation of the drive can rotate the hub **102** and the impeller **100** about a longitudinal axis or axis of rotation A-A' of the hub **102** within the flow device.

[0041] The impeller **100** can further include a plurality of blades **104** (also referred to as primary blades **104**, herein) extending radially outwards from the outer surface **102A** of the hub **102**. In one or more embodiments, each of the blades **104** can have a substantially curved profile, extending outwards from the hub **102** and curved in a predefined (clock-wise or counter-clock-wise) direction about the longitudinal axis or axis of rotation A-A' of the hub **102**, such that a plurality of fluidic passages **106** is formed or defined between the adjacent blades **104** to allow flow of the fluid therethrough. However, in other embodiments, the blades **104** can also have a substantially straight profile, without any limitations.

[0042] As illustrated, each of the blades **104** can include a leading edge **104A** located adjacent to the leading end **102-1** of the hub **102**, a trailing edge **104B** located adjacent to the trailing end **102-2** of the hub **102**, a coupling side extending between the trailing edge **104B** and the leading edge **104A** which remains attached to the outer surface **102A** of the hub **102**, a curved edge **104C** opposite to the coupling side which extends between the trailing edge **104B** and the leading edge **104A** of the corresponding blades **104**, and a tip T defining the trailing edge **104B**. The blades **104** can accordingly define a first face **104D** (surface) and a second face **104E** (opposite to the first face) between the leading edge **104A**, the trailing edge **104B**, the coupling side, and the curved edge **104C**.

[0043] In one or more embodiments, each of the blades **104** can be at least partially attached to the outer surface **102A** of the hub **102** such that at least a portion or length (L) of the tip end or trailing edge **104B** end of the corresponding blades **104** remains unattached to the outer surface **102A** of the hub **102**. Further, the unattached portion L of the blades **104** can extend at least partially beyond the outermost diameter (outer periphery of the trailing end **102-2**) of the hub **102**. Accordingly, the unattached portion L of the blades **104** can be free to deflect based on the rotational speed of the impeller **100** and correspondingly adjust the backsweep angle (β) of the corresponding blades **104** within a predefined range as shown in FIGS. 3A and 3B.

[0044] In one or more embodiments, each of the blades **104** can have a variable thickness between the coupling side and the curved edge **104C** of the corresponding blades **104**. However, in other embodiments, each of the blades **104** can also have a uniform thickness between the coupling side and the curved edge **104C** of the corresponding blades **104**. Further, in one or more embodiments, each of the blades **104** has a variable thickness between the leading edge **104A** and the trailing edge **104B** of the corresponding blades **104**. However, in other embodiments, each of the blades **104** can also have a uniform thickness between the leading edge **104A** and the trailing edge **104B** of the corresponding blades **104**.

[0045] In addition, in one or more embodiments, but not limited to the like, the width of each of the blades **104** can also be reduced while moving in a direction from the leading edge **104A** towards

the trailing edge **104B** of the corresponding blades **104**. However, in other embodiments, the width of each of the blades **104** can also be the same or may increase while moving between the leading edge **104A** and the trailing edge **104B** of the corresponding blades **104**.

[0046] In one or more embodiments, but not limited to the like, the thickness of each of the blades **104** can be reduced while moving in a radial direction from the coupling side towards the tip end or outer edge of the corresponding blades **104**. Further, in one or more embodiments, the thickness of each of the blades **104** can increase while moving in a direction from the leading edge **104A** toward the trailing edge **104B** of the corresponding blades **104**. Further, in some embodiments, the thickness of each of the blades **104** can first increase while moving from the leading edge **104A** towards a middle portion of the corresponding blades **104** and further reduce while moving towards the trailing edge **104B** of the corresponding blades **104**. However, in other embodiments, the thickness of each of the blades **104** can be reduced while moving in a direction from the leading edge **104A** toward the trailing edge **104B** of the corresponding blades **104**.

[0047] In one or more embodiments, as shown in FIG. 3A, when the impeller **100** is rotated at a lower rotational speed, the centrifugal force on the tip end or trailing edge **104B** of the blades **104** may be less. This reduction in centrifugal force may result in less deflection in the blades **104** of the impeller **100**, thereby maintaining a high backsweep angle (β). Further, in one or more embodiments, as shown in FIG. 3A, when the impeller **100** is rotated at higher rotational speeds, the increased centrifugal force may cause the blades **104** of the impeller **100** to straighten, leading to a decrease in the backsweep angle (β'). In one or more embodiments, the variable backsweep angle (β) to (β') design of the impeller **100** may allow it to operate in a rotational speed in a range of 60 to 105% of the design (nominal) speed. Accordingly, the backsweep angle (β) of the blades **104** of the impeller **100** may vary within the predefined range of 15 to 60 degrees from a radial direction from the hub **102**. Thus, this variation in the backsweep angle (β) in the blades **104** of the impeller **100** may improve the overall performance and operating range of the associated centrifugal compressor and the HVAC systems.

[0048] In one or more embodiments, the outer surface **102A** of the hub **102** can axially extend at the leading (upstream) end and then radially outward at the trailing (downstream) end such that an axial inlet to the longitudinal axis A-A' can be formed at the leading end **102-1** and a radial outlet can be formed at the trailing end **102-2** of the hub **102**, within a casing (not shown) or a shroud (shown in FIGS. 2A to 2C) associated with the flow device. As a result, the outer surface **102A** of the hub **102** can have a substantially curved profile, where the outer diameter of the hub **102** can increase in a direction while moving from the leading end **102-1** toward the trailing end **102-2** of the hub **102**. However, in other embodiments, the hub **102** can also have a cylindrical profile.

[0049] In one or more embodiments, a portion of the blades **104** near the leading end **102-1** can protrude generally radially outward from the hub **102** and another portion towards the trailing end **102-2** can protrude axially from the hub **102**. In one or more embodiments, the trailing edge **104B** of the blades **104** can be circumferentially offset from the corresponding leading edge **104A** of the blade **104**. Further, the portion, at the leading edge **104A**, of each blade **104** can be substantially curved in a direction, opposite to the predefined direction. Accordingly, the portion of the blades **104**, near the trailing end **102-2**, can remain generally axially oriented with respect to the radial outlet of the impeller **100**. However, the portion of the blades **104**, near the leading edge **104A**, can remain generally radially oriented or substantially parallel to a plane (normal to the rotational axis A-A') of the axial inlet of the impeller **100**.

[0050] In one or more embodiments, the impeller **100** can include a plurality of splitters **108** (also referred to as splitter blades or secondary blades, herein) configured in an alternating arrangement with the plurality of main blades **104** such that one of the splitters **108** remains between the adjacent main blades **104**. The splitters **108** can be configured to split the fluid flowing through the fluidic passages **106** formed between the adjacent blades **104**. The splitters **108** can also have a substantially curved profile that can extend radially outwards from the hub **102** and remain attached

to the outer surface **102A** of the hub **102**. Further, the length of the splitters **108** can be substantially smaller than the main blades **104**, such that a trailing edge of the splitters **108** can be free to deflect and can also extend beyond the outer diameter of the hub **102**. However, in one or more embodiments (not shown), the length of the splitters **108** can be substantially smaller than the main blades **104**, with the splitters **108** not extending beyond the outer diameter of the hub **102**.

[0051] Referring to FIGS. **2A** to **2C**, in one or more embodiments, the impeller **100** can include a shroud **110** encasing the hub **102**, the blades **104**, and the splitters **108**, such that a first end **110-1** of the shroud **110** located adjacent to the leading edge **104A** of the blades **104** or the leading end **102-1** of the impeller **100** remains open to form the axial inlet **110A** that allows inflow of a fluid therewithin. Further, a second end **110-2** of the shroud **110** located adjacent to the trailing edge **104B** of the blades **104** or trailing end **102-2** of the impeller **100** remains open to form the radial outlet **110B** that allows outflow of the fluid therefrom. Further, the shroud **110** can encase the hub **102** and the blades **104**, such that the unattached portion **L** of the blades **104** remains enclosed by and unattached to the shroud **110**. Furthermore, the unattached portion **L** of the blades **104** can extend at least partially out of the shroud **110** via the radial outlet.

[0052] In one or more embodiments, the shroud **110** can also have a substantially curved profile based on an outer profile of the hub **102** and the blades **104**. The diameter of the shroud **110** can increase while moving in a direction from the first end **110-1** toward the second end **110-2** of the shroud **110** or from the leading end **102-1** towards the trailing end **102-2** of the hub **102**. As illustrated, the axial inlet **110A** of the shroud **110** or impeller **100** can have an opening of a predefined diameter extending along a plane that is normal to the longitudinal axis **A-A'** at the leading end **102-1**. Further, the radial outlet **110B** of the shroud **110** or impeller **100** can have an opening of a predefined width that remains axially oriented and extends circumferentially around an outer diameter of the trailing end **102-2** of the hub **102**.

[0053] In one or more embodiments, in a shrouded impeller **100** as shown in FIG. **2A** to **2C**, the integral shroud **110** can surround the blades **104** so that the blades **104** extend across a flow path between the surface of the hub **102** and the surface of the shroud **110**. The shroud **110** can rigidify the blades **104** and reduce vibrations. This may allow the use of a relatively thinner blade structure than in an equivalent open impeller **100** wherein the blades **104** are secured only to the hub **102**. Further, in one or more embodiments, in an open impeller **100** design (without the shroud **110**) as shown in FIG. **1A** to **1C**, the edges of the blades **104** can closely interface with a separate casing on the compressor. There may be position control between the impeller **100** and casing to balance the loss of efficiency due to leakage against the risk of damage from impeller-to-casing contact.

[0054] Referring to FIGS. **1A** to **2C**, in one or more embodiments, the blades **104**, the hub **102**, the splitters **108**, and the shroud **110** of the impeller **100** can be made of a non-metallic, non-corrosive material. Further, the blades **104**, the hub **102**, the splitters **108**, and the shroud **110** of the impeller **100** can be made of alloys. However, in other embodiments, the flow device and the impeller **100** can be constructed using any standard or newly developed materials and methods, and all such embodiments are well within the scope of the subject disclosure. In one or more embodiments, the impeller **100** or the components of the impeller **100** may be manufactured using additive manufacturing processes and/or 3D printing techniques. However, the impeller **100** or its components can also be manufactured using any standard or newly developed methods, and all such embodiments are well within the scope of the subject disclosure.

[0055] Further, in one or more embodiments, the plurality of blades **104** can be specifically made of a composite material having a predetermined flex modulus, which may be coated with layer(s) of non-corrosive material(s). This may provide the blades **104** with intrinsic flexibility, which allows the blades **104** to deflect and adjust the backsweep angle based on the rotational speed of the impeller **100**.

[0056] Thus, the subject disclosure overcomes the challenges associated with existing impellers, by

providing an improved impeller that is capable of adjusting its backsweep angle in response to changing operational speed without any external control mechanism. It is to be appreciated that when the compressor (flow device) is operating at a lower rotational speed, typical of a part-load condition in HVAC systems (chillers or heat pumps), the centrifugal force is less. This reduction in force results in less deflection in the blades of the impeller, thereby preserving the high backsweep angle. Further, when the compressor is operating at higher rotational speeds, which are common during full-load conditions in HVAC systems (chillers or heat pumps), the increased centrifugal force causes the blades of the impeller to straighten, leading to a decrease in the backsweep angle. This improves the overall performance and operating range of the associated centrifugal compressor and the HVAC systems.

[0057] While the subject disclosure has been described with reference to exemplary embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the subject disclosure as defined by the appended claims. Modifications may be made to adopt a particular situation or material to the teachings of the subject disclosure without departing from the scope thereof. Therefore, it is intended that the subject disclosure not be limited to the particular embodiment disclosed, but that the subject disclosure includes all embodiments falling within the scope of the subject disclosure as defined by the appended claims.

[0058] In interpreting the specification, all terms should be interpreted in the broadest possible manner consistent with the context. In particular, the terms “comprises” and “comprising” should be interpreted as referring to elements, components, or steps in a non-exclusive manner, indicating that the referenced elements, components, or steps may be present, or utilized, or combined with other elements, components, or steps that are not expressly referenced. Where the specification claims refer to at least one of something selected from the group consisting of A, B, C . . . and N, the text should be interpreted as requiring only one element from the group, not A plus N, or B plus N, etc.

Claims

1. An impeller for a flow device, the impeller comprising: a hub; and a plurality of blades extending radially outwards from the hub, wherein each of the blades is at least partially attached to an outer surface of the hub such that at least a portion of a tip end, opposite to the hub, of the corresponding blades remains unattached to the outer surface of the hub, wherein the unattached portion of the blades is free to deflect based on a rotational speed of the impeller and correspondingly adjust a backsweep angle of the corresponding blades within a predefined range.
2. The impeller of claim 1, wherein the unattached portion of the blades extends at least partially beyond an outermost diameter of the hub.
3. The impeller of claim 1, wherein the predefined range of the backsweep angle ranges from 15 to 60 degrees from a radial direction from the hub.
4. The impeller of claim 1, wherein each of the blades has a substantially curved profile, extending in a predefined direction about a longitudinal axis or axis of rotation of the hub, such that a plurality of fluidic passages is formed or defined between adjacent blades in the plurality of blades to allow flow of a fluid therethrough.
5. The impeller of claim 1, wherein the outer surface of the hub has a substantially curved profile, wherein an outer diameter of the hub increases in a direction while moving from a leading end towards a trailing end of the hub, wherein the unattached portion of the hub extends at least partially beyond the outer diameter of the trailing end of the hub.
6. The impeller of claim 1, wherein each of the blades comprises a coupling side attached to the outer surface of the hub, a leading edge located adjacent to a leading end of the hub, a trailing edge located adjacent to a trailing end of the hub, and a curved edge opposite to the coupling side and

- extending between the trailing edge and the leading edge of the corresponding blades.
- 7.** The impeller of claim 1, wherein a trailing edge of the plurality of blades are circumferentially offset from a corresponding leading edge of the blade.
- 8.** The impeller of claim 1, wherein a portion, at a leading edge, of each of the blades is substantially curved in a direction opposite to a predefined direction.
- 9.** The impeller of claim 6, wherein each of the blades has a variable thickness between the leading edge and the trailing edge of the corresponding blades.
- 10.** The impeller of claim 9, wherein the thickness of each of the blades reduces while moving in a radial direction from the coupling side towards the tip end or outer edge of the corresponding blades.
- 11.** The impeller of claim 6, wherein each of the blades has a variable thickness between the coupling side and the curved edge of the corresponding blades.
- 12.** The impeller of claim 6, wherein the thickness of each of the blades reduces while moving in a direction from the leading edge towards the trailing edge of the corresponding blades.
- 13.** The impeller of claim 6, wherein the thickness of each of the blades increases while moving from the leading edge towards a middle portion of the corresponding blades and further reduces while moving towards the trailing edge of the corresponding blades.
- 14.** The impeller of claim 1, wherein the impeller comprises a shroud encasing the hub and the plurality of blades, such that a first end of the shroud located adjacent to a leading edge of the blades remains open to allow inflow of a fluid therewithin, and a second end of the shroud located adjacent to a trailing edge of the plurality of blades remains open to allow outflow of the fluid therefrom.
- 15.** The impeller of claim 14, wherein the shroud encases the hub and the plurality of blades, such that the unattached portion of the blades remains enclosed by and unattached to the shroud.
- 16.** The impeller of claim 14, wherein the shroud has a substantially curved profile, wherein a diameter of the shroud increases while moving in a direction from a leading end towards a trailing end of the hub.
- 17.** The impeller of claim 1, wherein the plurality of blades, the hub, and a shroud of the impeller are made of a non-metallic material.
- 18.** The impeller of claim 1, wherein the plurality of blades is made of a composite material having a predetermined flex modulus.
- 19.** The impeller of claim 1, wherein the impeller comprises a plurality of splitters configured in an alternating arrangement with the plurality of blades such that one of the splitters remains between the adjacent blades, wherein the splitters extend radially outwards from the hub such that at least a portion of a tip end, opposite to the hub, of the corresponding splitter remains unattached to the outer surface of the hub.
- 20.** The impeller of claim 1, wherein the flow device is selected from a group comprising a centrifugal compressor, a centrifugal pump, and a turbine.
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